

Bearing and shortest distance

Maps used in navigation, for example, may require equal distances or constant bearings. *Bearing* is a way to express direction as an angle measured from a meridian in relation to the North position (0°). A constant bearing is maintained when you travel along a *rhumb line* (also called a *loxodrome*). Any straight lines between two points in a Mercator cylindrical projection are rhumb lines, since they maintain a constant angle in relation to the meridians.

There is no such thing as a straight line on the surface of a sphere or ellipsoid. The *shortest distance* between two points and the *line of constant bearing* are both curves.

The shortest path between two points on the surface of a sphere is the *arc of a great circle*. The *great circle* (also called *orthodrome*) is any circle dividing a sphere into two equal hemispheres. The equator and meridians are both rhumb lines and great circles, but if you travel along any oblique great circle, the bearing changes at every point on the line. Great circle arcs appear as curves in a Mercator projection but are projected as straight lines in a Gnomonic projection, where rhumb lines are curves.

Figure 1 compares rhumb lines and great circles showing both paths between London and Los Angeles (an actual flight path is usually chosen somewhere in between them, since it depends on factors like flight regulations, winds, climate and politics.)

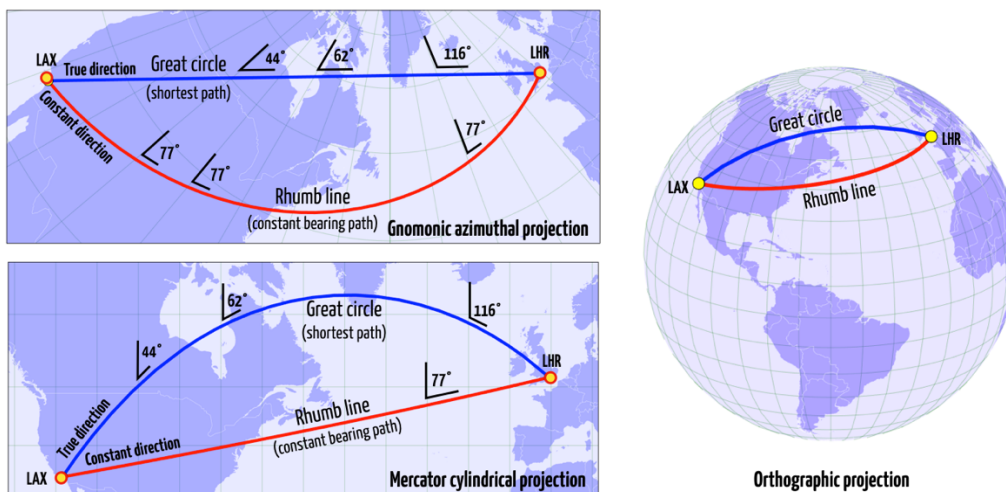


Figure 1 – A path between two points on the surface of a sphere is represented differently, depending on the projection. Code: [Measuring/2-loxodrome.html](https://github.com/d3/d3-geo-projection/blob/master/src/measure.js)

Straight lines in D3 are automatically drawn as great circles. If you need to draw a rhumb line between two points, you need to resample the GeoJSON object, adding intermediate points so it can be interpolated. See [Measuring/2-loxodrome.html](https://github.com/d3/d3-geo-projection/blob/master/src/measure.js) which uses the `resample()` function available from github.com/d3/d3-geo-projection/issues/75 (by Mike Bostock).